



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
Mid-Semester Examination-Autumn 2025

Subject: Discrete Mathematics
Full Marks: 25

Subject Code(s): COM 2101/COM 4102
Time Allotted: 2 h

Answer any 10 Questions

$$2.5 \times 10 = 25$$

1. How many non-zero positive integers less than equal to 2000 are divisible by either 9, 11 or 13?
2. Let, A be some fixed 10-element subset of $\{1, 2, 3, 4, \dots, 50\}$. Prove that, A possesses two different 5-element subsets, the sum of whose elements are equal.
3. Can you find out a number N such that $N!$ (N factorial) has 11 trailing zeros at the end. Give reasons in support of your answer.
4. How many factors of $2^5 \cdot 3^6 \cdot 5^2$ are perfect squares?
5. Consider the sequence $\langle S_n \rangle$ given by $S_0 = 1$ and $S_1 = -3$ and $S_n = 7S_{n-1} - 12S_{n-2}$, for $n \geq 2$. Find a closed form solution for S_n .
6. Let, S be the formula : $((p \Rightarrow q) \text{ and } (q \Rightarrow r) \text{ and } p) \Rightarrow r$. Show that it's a tautology.
7. Find the CNF of the propositional formula $((p \text{ and } q) \Rightarrow r) \text{ and } (p \Rightarrow q)$.
8. Find out the truth values of the following predicates (give reasons) -
 - a. $\forall m \exists n \{n > 2^m\}$
 - b. $\exists y \forall x \{x + y = 0\}$
 - c. $\exists y \forall x \{xy = 0\}$
9. In how many way, can we write 9 as a sum of three positive integers where each summand is at most 4?

- ✓ 10. Show that $((a \wedge b) \longrightarrow c) \iff ((a \longrightarrow c) \vee (b \longrightarrow c))$.
- ✓ 11. Show that from any three integers, one can always choose two integers a and b , so that $a^3b - ab^3$ is divisible by 10.
- ✗ 12. Find out three triplets (x, y, z) of positive integers x, y and z ($x, y, z > 0$) such that $(x+y+z) + (xy + yz + zx) + xyz = 179$. How many such triplets exist?